Chapter 39

Exam pocket cards

the handful of facts that unlock each topic — like y = mx + c for lines

You said it yourself: once **y = mx + c**, the gradient formula, and "parallel / perpendicular" are in your head, every straight-line question is rinse and repeat. These cards do that for every exam-critical topic.

How to use them

- Read the matching chapter once for the why (pictures, derivations).
- Memorise the card until you can write it from blank paper.
- Drill past-paper questions of that type until the card fires automatically.
- Tier 1 cards first (pass → 70%). Tier 2 after those are automatic.

Past papers reviewed: Goldsmiths Foundation Maths IS50002 (2010–2019), NCC FDM May 2016, Bolton EEE4011, plus your NTU lecture sequence L01–L17. **IS51002** (graph theory, relations, induction) is a **different** module — not the focus of this guide.

Four-week priority map

Your NTU module end-goals (graphs, descriptive stats, rearrange equations, calculus on curves, geometry & probability, vectors) sit mostly in Tier 1. Goldsmiths-style foundation papers match Tier 1 almost question-for-question.

Week 1 — number fluency

Ch 1–5, 9–10: fractions, decimals, %, ratio, indices, bases. No calculator habits.

Week 2 — algebra engine

Ch 8, 11–15: expand, factorise, transpose, linear / simultaneous / quadratic.

Week 3 — structures + chance

Ch 6–7, 16–18, 25: sets, logic, matrices, sequences, functions, probability trees.

Week 4 — graphs, geo, calc core

Ch 19–20, 24, 28, 32, 34, 36: lines, stats, trig, vectors, differentiate, integrate.

Daily drill: **one pocket card from memory**, then **two past-paper parts** of that type. Aim for speed with full working shown — foundation papers often ban calculators.

What the papers actually ask

Goldsmiths IS50002 Foundation Maths is remarkably stable across years. Ten questions, ~10 marks each, no calculator. Typical map → guide chapters:

Q-type (every year-ish)	Guide chapters
Fractions, %, order of ops, d.p./s.f., standard form	**2, 3, 4, 9**
Algebra: expand, substitute, indices, factors	**8, 9, 11, 12**
Number bases (bin/hex) ± matrices sometimes	**10, 16**
Sets, Venn, truth tables (tautology / contingency / contradiction)	**6, 7**
Linear, simultaneous, quadratic	**15**
Sequences and series (AP/GP, Σ)	**17**
Functions: evaluate, compose, inverse	**18**
Graphs / straight lines / intersection	**19, 20**
Trig: Pythagoras, sin/cos/tan, sketch sin	**27, 28, 29**
Matrices 2×2: +, ×, I, inverse	**16**
Probability: dice/coins, trees, with/without replacement	**25**

NTU lectures add descriptive stats (L04), charts (L05), Bayes/Normal (L08–09), vectors (L10), polar (L11), and calculus (L14–17). Those are Tier 1 for your course even when Goldsmiths papers stop before them.

Skip for now (unless your paper is IS51002-shaped): graph theory, relations / orders, proof by induction, homogeneous coordinates, complex numbers, ODEs.

Card index

Tier 1 first. Tick each card when you can reproduce it blank.

○	Ch 02	Fractions	T1
○	Ch 03	Decimals, rounding, standard form	T1
○	Ch 04	Percentage and ratio	T1
○	Ch 09	Indices	T1
○	Ch 10	Number bases	T1
○	Ch 12	Expand and factorise	T1
○	Ch 14	Transposing	T1
○	Ch 15	Solving equations	T1
○	Ch 06	Sets and Venn	T1
○	Ch 07	Logic / truth tables	T1
○	Ch 17	Sequences and series	T1
○	Ch 18	Functions	T1
○	Ch 20	Straight line	T1
○	Ch 16	Matrices (2×2)	T1
○	Ch 25	Probability	T1
○	Ch 28	Trigonometry (right-angled)	T1
○	Ch 24	Statistics (descriptive)	T1
○	Ch 32	Vectors	T1
○	Ch 34	Differentiation (core)	T1
○	Ch 36	Integration (core)	T1
○	Ch 22	Logs and exponentials	T2
○	Ch 26	Bayes and Normal	T2
○	Ch 35	Further differentiation	T2
○	Ch 33	Trig identities / polar	T2

TIER 1 · exam core

FRACTIONS

full explanation → Chapter 2: Fractions

Add / subtract

$$\frac{a}{b} \pm \frac{c}{d} = \frac{ad \pm bc}{bd}$$

common denom is already in the formula

Multiply

$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}$$

Divide

$$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$$

flip the second, then multiply

Inputs

whole = n/1; mixed → improper first; simplify only if asked

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

DECIMALS, ROUNDING, STANDARD FORM

full explanation → Chapter 3: Decimals

d.p.

look at the next digit: ≥5 rounds up

s.f.

start counting from the first non-zero digit

Standard form

$$a \times 10^n,\ 1 \leq a < 10$$

scientific notation is the same idea

Recurring → fraction

x = 0.ȧḃ…; multiply by 10^k; subtract; solve for x

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

PERCENTAGE AND RATIO

full explanation → Chapter 4: Percentage and ratio

Percent of

$$\frac{x}{100} \times y$$

Multiplier

$$\text{new} = \text{old} \times \frac{100 \pm r}{100}$$

\+ for increase, − for decrease

Reverse %

$$\text{original} = \frac{\text{new}}{1 \pm r/100}$$

undo a completed % change

Share in ratio a:b:c

parts = a+b+c; one part = total ÷ parts; each = parts × one

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

INDICES

full explanation → Chapter 9: Indices (powers)

Laws

$$a^m a^n = a^{m+n},\ \frac{a^m}{a^n} = a^{m-n},\ (a^m)^n = a^{mn}$$

Zero / negative

$$a^0 = 1,\ a^{-n} = \frac{1}{a^n}$$

$a \neq 0$

Fractional

$$a^{1/n} = \sqrt[n]{a},\ a^{m/n} = (\sqrt[n]{a})^m$$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

NUMBER BASES

full explanation → Chapter 10: Number bases

Binary → decimal

add place values: … 16 8 4 2 1

Decimal → binary

repeated ÷2; remainders bottom→top

Hex

groups of 4 bits; A=10 … F=15

Binary +/−

same as decimal but place values are powers of 2; show working

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

EXPAND AND FACTORISE

full explanation → Chapter 12: Factorisation

Expand

$$(a+b)(c+d) = ac + ad + bc + bd$$

every left term hits every right term

Common factor

$$ab + ac = a(b+c)$$

pull out the HCF of the terms

Quadratic

find two numbers that multiply to ac and add to b; split middle

Difference of squares

$$a^2 - b^2 = (a-b)(a+b)$$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

TRANSPOSING

full explanation → Chapter 14: Transposing formulae

Balance

do the same inverse op to both sides

Undo order

unwrap outside-in (opposite of BIDMAS)

Goal

get the wanted letter alone on one side

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

SOLVING EQUATIONS

full explanation → Chapter 15: Solving equations

Linear

expand → collect like terms → divide

Simultaneous

elimination (align one letter) or substitution

Quadratic factorise

$$(x-p)(x-q)=0 \Rightarrow x=p \text{ or } x=q$$

zero-product rule

Quadratic formula

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

when factorising fails

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

SETS AND VENN

full explanation → Chapter 6: Sets

Symbols

∪ union (OR), ∩ intersection (AND), ′ complement, − difference

Count union

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

don't double-count the overlap

Power set

$$|P(S)| = 2^n$$

all subsets of an n-element set

Venn habit

fill from the middle out; unknown region = x

TIER 1 · exam core

LOGIC / TRUTH TABLES

full explanation → Chapter 7: Logic

Rows

$$2^n$$

n letters → that many rows

AND / OR / NOT

∧ both T; ∨ either T; ¬ flips

Implication

$$P \Rightarrow Q$$

false only when P is T and Q is F

Classify

final column: all T = tautology; all F = contradiction; mixed = contingency

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

SEQUENCES AND SERIES

full explanation → Chapter 17: Sequences and series

AP nth term

$$a_n = a + (n-1)d$$

a = first term, d = common difference

AP sum

$$S_n = \frac{n}{2}(2a + (n-1)d) = \frac{n}{2}(a + l)$$

l = last term

GP nth term

$$a_n = ar^{n-1}$$

r = common ratio

GP sum

$$S_n = \frac{a(1-r^n)}{1-r} \ (r \neq 1)$$

|r|<1:\ S_\infty=\frac{a}{1-r}

cover the card - write from memory - check - then do 2 paper parts

TIER 1 · exam core

FUNCTIONS

full explanation → Chapter 18: Functions

Evaluate

replace every x with the given number; brackets help

Compose

$$(f \circ g)(x) = f(g(x))$$

inside first

Inverse

swap x and y (or set y = f(x)), then make x the subject; rename

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

STRAIGHT LINE

full explanation → Chapter 20: The straight line

Core

$$y = mx + c$$

m = gradient, c = y-intercept

Gradient

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

rise over run

Point-gradient

$$y - y_1 = m(x - x_1)$$

when you know a point and m

Parallel / perp

$$m_1 = m_2; \quad m_1 m_2 = -1$$

perp: negative reciprocal

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

MATRICES (2×2)

full explanation → Chapter 16: Matrices

Multiply

row of left · column of right; sizes must match

Determinant

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc$$

Inverse

$$\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

swap diagonal, flip off-diagonal signs

Solve Ax = b

x = $A^{-1}b$ when det ≠ 0

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

PROBABILITY

full explanation → Chapter 25: Probability and Bayes

Basic

$$p = \frac{\text{favourable}}{\text{total}}$$

equally likely outcomes

AND / OR

independent: multiply; mutually exclusive: add

Conditional

$$p(E|F) = \frac{p(E \cap F)}{p(F)}$$

Tree

multiply along a path; add across paths; watch with/without replacement

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

TRIGONOMETRY (RIGHT-ANGLED)

full explanation → Chapter 28: Intro to trigonometry

SOH CAH TOA

$$\sin = \frac{o}{h},\ \cos = \frac{a}{h},\ \tan = \frac{o}{a}$$

Pythagoras

$$h^2 = o^2 + a^2$$

Exact values

0°, 30°, 45°, 60°, 90°: memorise sin/cos/tan table

CAST / graph

find all solutions in the given range from the principal angle

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

STATISTICS (DESCRIPTIVE)

full explanation → Chapter 24: Statistics

Mean

$$\bar{x} = \frac{\sum x}{n}$$

or Σfx / Σf from a frequency table

Median / mode

middle when ordered; most frequent

IQR

Q3 − Q1; resistant to outliers

SD

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} \text{ or } n - 1$$

spread around the mean

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

VECTORS

full explanation → Chapter 32: Vectors

Add / scale

component-wise; k⟨x,y⟩ = ⟨kx, ky⟩

Magnitude

$$|\mathbf{a}| = \sqrt{x^2 + y^2 (+z^2)}$$

Dot product

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos\theta = x_1 x_2 + y_1 y_2$$

perp ⇒ dot = 0

Unit vector

$$\hat{\mathbf{a}} = \mathbf{a}/|\mathbf{a}|$$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

DIFFERENTIATION (CORE)

full explanation → Chapter 34: Gradients of curves

Power rule

$$\frac{d}{dx}x^n = nx^{n-1}$$

constants differentiate to 0

Sum / multiple

term by term; constant factors stay

Turning points

dy/dx = 0; 2nd derivative (or sign check) for max/min

Tangent

m = dy/dx at the point; then $y - y_1 = m(x - x_1)$

cover the card · write from memory · check · then do 2 paper parts

TIER 1 · exam core

INTEGRATION (CORE)

full explanation → Chapter 36: Integration and areas

Reverse power

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1)$$

Definite

$$\int_a^b f = F(b) - F(a)$$

area under curve (watch sign below axis)

1/x and e^x

$$\int \frac{1}{x} = \ln|x| + C, \quad \int e^x = e^x + C$$

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

LOGS AND EXPONENTIALS

full explanation → Chapter 22: Logarithms

Definition

$$y = \log_b x \iff b^y = x$$

Laws

$\log(ab) = \log a + \log b,\ \log(a/b) = \log a - \log b,\ \log(a^k) = k \log a$

Solve a^x = b

take logs both sides, or rewrite as log

e and ln

ln = log base e; inverse of exp

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

BAYES AND NORMAL

full explanation → Chapter 26: The Normal distribution

Bayes

$$p(A|B) = \frac{p(B|A)\,p(A)}{p(B)}$$

prior → posterior after evidence

Total probability

partition B via A and A′ (or a tree)

z-score

$$z = \frac{x - \mu}{\sigma}$$

then use the Normal table

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

FURTHER DIFFERENTIATION

full explanation → Chapter 35: Chain, product, quotient

Product

$$(uv)' = u'v + uv'$$

Quotient

$$(u/v)' = (u'v - uv')/v^2$$

Chain

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

function of a function

cover the card · write from memory · check · then do 2 paper parts

TIER 2 · stretch / later lectures

TRIG IDENTITIES / POLAR

full explanation → Chapter 33: Polar coordinates

Pythagorean

$$\sin^2\theta + \cos^2\theta = 1$$

Polar ↔ Cartesian

$$x = r\cos\theta,\ \ y = r\sin\theta,\ \ r = \sqrt{x^2 + y^2}$$

fix θ for quadrant

cover the card · write from memory · check · then do 2 paper parts